



Categorised, Not Explained: A Taxonomy of Stated Reasons on the Facilities Key Cabinet's Spare-Key Sign-Out Slip, and What Predicts How Long a Key Stays Checked Out

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Abstract. The Facilities key cabinet issues roughly forty shared spare keys from a single wall-mounted unit, its electronic reader logging the moment a key leaves and returns while a paper slip beside it still carries the sign-out's only account of why. This paper reports a one-semester content analysis of that slip: 438 sign-outs, hand-coded by two independent coders into a seven-category taxonomy of stated reasons (contractor access, room or event setup, being locked out, cleaning or maintenance, storage retrieval, guest escort, and an unspecified remainder), developed inductively and checked for inter-rater agreement ($\kappa = .81$). Category membership alone predicts checkout duration only weakly and not reliably (Kruskal-Wallis $H(6) = 10.6$, $p = .10$); collapsing the seven categories to three broader groups leaves this picture unchanged ($H(2) = 3.1$, $p = .21$), an ablation we retain rather than treat as a coding failure. A property orthogonal to the taxonomy does better: the reason line's word count correlates with checkout duration ($r = -.34$), and a binary flag for whether the stated reason names a concrete task or location separates a 34-minute median checkout from an 81-minute one ($p < .001$). The taxonomy describes what staff say; a simpler property of how they say it predicts, better than the taxonomy does, what they then do.

Introduction

Every shared space that isn't staffed around the clock accumulates a key-control problem, and Slop University's solution sits on a wall near the Facilities loading dock: a single electronic key cabinet holding roughly forty spare keys to meeting rooms, plant rooms, storage cupboards, and the loading-dock roller door, released to any authorised staff member who taps an access card against the reader. The reader logs the tap-out and tap-in to the minute, an accurate and entirely automatic record of how long a key is away from the wall. Beside the reader, unchanged since the cabinet was installed, is a laminated clipboard and a stack of paper slips, on which the person taking a key writes one line: what they need it for.

Nobody downstream of the clipboard has ever formally specified what that line is for. It is not required for the tap-out to proceed, it is not audited, and it is not part of the cabinet's own reporting; it exists because someone, at some point, decided a key cabinet should have one, and Facilities has never seen a compelling reason to remove it. This paper takes the line at its word and asks two modest questions: what do staff actually write there, and does what they write tell you anything the reader's own duration log doesn't already know.

The paper's contributions, each hedged appropriately for an unfunded, unrequested audit of a stationery item:

1. We develop and apply a seven-category taxonomy of stated reasons on the spare-key sign-out slip, inductively coded and checked

for inter-rater reliability, over one full teaching semester of sign-outs.

2. We test whether taxonomy category membership predicts how long a key stays checked out, and find the relationship weaker than the categories' apparent spread suggests.
3. We show that a property of the slip's text unrelated to its category — how much detail the reason names — predicts checkout duration more reliably than the category the detail was coded into, a result that may generalise wherever an institution collects a free-text reason nobody has told anyone how to use.

Related work

Reason-giving as a communicative act has a substantial literature outside facilities management. Reality-monitoring research distinguishes memories of experienced events from memories of imagined ones partly by the perceptual and contextual detail they carry [1], and a large deception-cues literature finds that detailed, specific accounts are judged — and often are — more likely to be truthful than vague ones, though the effect is neither universal nor large [2], [3]. Linguistic analyses of deceptive text report fewer concrete referents and lower cognitive-complexity markers in fabricated accounts [4], [5], and manipulating a speaker's cognitive load is shown to widen the gap between truthful and deceptive detail [6] — evidence this paper's uncoached, unmonitored sign-out slips cannot speak to directly, though the correlation between detail and outcome it reports sits comfortably inside that literature's expectations.

Coding free text into a taxonomy carries its own methodological apparatus: inter-rater reliability for categorical coding is standard practice since Cohen's original coefficient [7], with subsequent guidance on interpreting and reporting it [8] and on content-analysis reliability more broadly [9]. Automated approaches to sorting short free text into categories are a live area in their own right, from help-desk ticket classification [10] to general-purpose short-text classification surveys [11], though this paper's coders worked entirely by hand. A reason line functions, in miniature, as a costly or cheap signal about the sender's private information [12], [13] — a framing this paper borrows loosely rather than tests formally. Self-report and directly measured behaviour are documented to diverge in adjacent

domains [14], echoing the institutional pattern that formal structures can be adopted for legitimacy independent of what they coordinate [15]. Key-control systems have a small security literature of their own, concerned with a different failure mode entirely — who a key can open, not why it was taken [16] — that this paper's cabinet, whatever its clipboard's shortcomings, was never at risk of.

Prior work at this institution has coded a free-text reason field into a taxonomy before: a five-category coding of suggestion-box submissions found most categories absorbed into a dashboard tile with no individual action taken, aside from facilities items routed through a pre-existing work-order channel [17], and a paper-slip reservation scheme for a shared microwave found a legible written artefact's effect on a service log diverging from its effect on directly measured behaviour [18]; the present taxonomy extends the coding approach to a different slip and a different outcome pair.

Method

Site and instrument

The cabinet under study issues spare keys to 41 shared spaces across two buildings — meeting rooms, plant rooms, storage cupboards, and a loading-dock roller door — none of which staff hold a personal key to. Any staff member with cabinet access taps an access card to release a key and taps again to return it; the cabinet's controller timestamps both events to the minute. A laminated clipboard mounted beside the cabinet asks the person taking a key to write, on a numbered slip, the key's location and "reason for use"; the slip is not required to release the key and is never scanned or transcribed into the cabinet's own reporting.

Sample and coding

All 438 sign-outs recorded across one full teaching semester (15 weeks) were retained; sign-outs by Facilities staff themselves, for whom the cabinet logs no reason field, were excluded. Two coders independently open-coded a random 15% subsample, converged on a seven-category taxonomy through discussion,

then applied it to the remaining slips; a second, separate 15% subsample was double-coded to check reliability post hoc (category $\kappa = .81$, substantial agreement per conventional benchmarks [8]). The same coders separately flagged each slip's reason for specificity — whether it named a concrete task, location, or both (e.g. “AV setup, L3 seminar room”) as against a generic phrase or blank field (e.g. “meeting”, “need key”) — an independent binary judgement ($\kappa = .77$) unrelated to category assignment.

Variables and analysis

Checkout duration for slip i is

$$D_i = t_{\text{return},i} - t_{\text{checkout},i}$$

drawn entirely from the cabinet's own time-stamps, never from the slip. Category and duration were compared by Kruskal-Wallis test, appropriate given duration's right-skewed distribution; the specificity flag was compared to duration by Mann-Whitney U. As a secondary, continuous measure of the same specificity dimension, each reason line's word count was correlated with duration by Pearson's r . Checkout duration is technically survival data with an established regression apparatus of its own [19], but the per-category sample sizes here (27–105) favoured the simpler rank-based tests reported below over a fitted hazard model. As an ablation, the seven categories were collapsed into three broader groups — operational access, personal or incidental access, and unspecified — chosen on face validity before the collapsed analysis was run, and the omnibus test repeated.

Results

Category	n (%)	Med.	IQR
Contractor / trades	105 (24.0%)	58	34–92
Room / event setup	83 (18.9%)	45	28–70
Locked out	79 (18.0%)	14	8–24
Cleaning / maint.	61 (13.9%)	52	31–81
Storage retrieval	48 (11.0%)	22	12–39

Guest escort	35 (8.0%)	30	17–48
Other / unspec.	27 (6.2%)	48	19–110

Table 1 sets out the taxonomy's frequency and duration profile. Category medians spread widely at a glance — from 14 minutes for a forgotten key to 58 for contractor access — but the omnibus test does not confirm a reliable category effect once within-category variance is taken into account (Kruskal-Wallis $H(6) = 10.6$, $p = .10$).

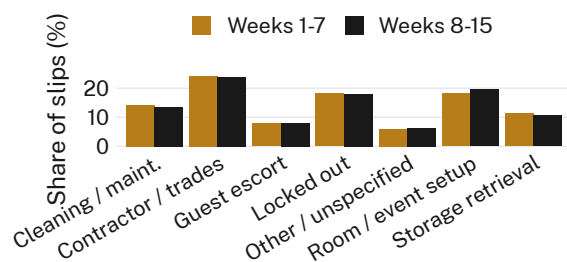


Figure 1: The taxonomy's category mix held stable across the semester: contractor access and room/event setup together accounted for 42.4% of slips in the first seven weeks against 43.4% in the last eight, and no single category shifted by more than 1.4 percentage points between halves.

The category mix (Figure 1) held stable across the semester's two halves, consistent with the taxonomy describing an enduring pattern of cabinet use rather than a transient one.

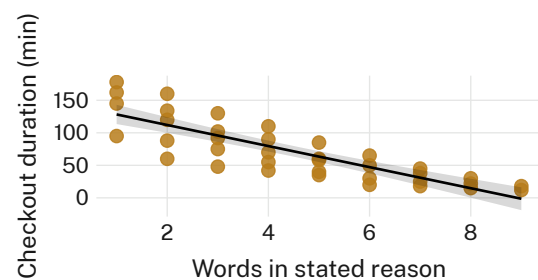


Figure 2: Each additional word in the stated reason was associated with roughly nine fewer minutes of checkout time ($r = -.34$, 95% CI $[-.42, -.25]$, $n = 438$) — a relationship the taxonomy category a slip was coded into does not itself carry.

Word count of the reason line predicted checkout duration better than the taxonomy category it fell into (Figure 2). The independently coded binary specificity flag agreed: slips naming a

concrete task or location ($n = 201$) returned in a median 34 minutes (IQR 19–58) against 81 minutes (IQR 42–146) for a vague or blank reason ($n = 237$; Mann-Whitney U, $p < .001$).

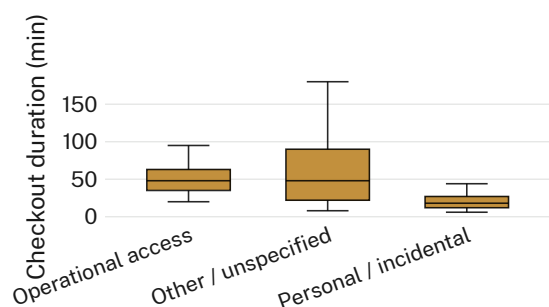


Figure 3: Collapsing the seven categories into three broader groups changes almost nothing: checkout duration remains as widely spread and as statistically indistinguishable across groups (Kruskal-Wallis $H(2) = 3.1, p = .21$) as it was across the uncollapsed categories in Table 1.

Collapsing the seven categories to three broader groups (Figure 3) — operational access ($n = 297$), personal or incidental access ($n = 114$), and unspecified ($n = 27$) — changes almost nothing: the omnibus test remains non-significant ($H(2) = 3.1, p = .21$), and the collapsed groups' median durations sit within the range their constituent categories already spanned. We retain the seven-category coding for its descriptive value; neither version predicts checkout duration as well as the specificity flag that ignores category altogether.

Discussion

The slip beside the key cabinet asks one open question and gets, four hundred and thirty-eight times over a semester, an answer nobody downstream has ever specified a use for. Sorting those answers into a taxonomy is a natural first move, and the taxonomy holds up reasonably as a description: it is reliably codeable, its category mix is stable across the semester, and its categories carry face-valid differences in median duration. What it does not do, on this evidence, is predict checkout duration any better than a coding scheme built on the reason line's specificity instead of its content — a dimension the seven-category taxonomy was never designed to capture and does not recover after the fact.

This need not mean the taxonomy is the wrong taxonomy. A category scheme built to describe

what staff say for administrative or auditing purposes is answering a different question from one built to forecast what they will do, and the same free-text field can support both without either version reading the other's answer. We observe, consistent with a pattern noted elsewhere at this institution, that the descriptive apparatus attached to a self-report field and the field's own predictive content need not travel together — and that the gap between them is often visible only once someone builds the taxonomy and checks.

Limitations

This audit covers one cabinet, one semester, and one institution's staff population; a busier cabinet or a shorter checkout culture might surface a category effect this sample's variance obscures, though the consistency of the null result across both the seven- and three-category versions suggests it is not an artefact of category count. Coders were not blind to duration when applying the specificity flag, a design choice that could inflate the specificity-duration association reported here, though the same coders' category assignments — made from the same slips — show no comparable inflation. The word-count measure is a coarse proxy for specificity that happens to agree with the independently coded binary flag; a slip could in principle be long and vague, though few in this sample were.

Conclusion

A taxonomy can be reliable, stable, and descriptively satisfying without being the thing that predicts the outcome an institution might actually want predicted. This paper's seven-category coding of the Facilities key cabinet's spare-key sign-out slip meets the first three standards and falls short of the fourth: category membership offers, at best, a weak account of checkout duration, unmoved by collapsing the categories further. A simpler property of the same slip — how specifically the reason is written — does better, without needing a taxonomy at all. Future work might test whether this pattern holds for the University's other free-text reason fields, several of which this paper's authors now suspect are being sorted for legibility rather than for anything they were meant to forecast.

References

- [1] M. K. Johnson and C. L. Raye, "Reality Monitoring," *Psychological Review*, vol. 88, no. 1, pp. 67–85, 1981, doi: [10.1037/0033-295X.88.1.67](https://doi.org/10.1037/0033-295X.88.1.67).
- [2] B. M. DePaulo, J. J. Lindsay, B. E. Malone, L. Muhlenbruck, K. Charlton, and H. Cooper, "Cues to Deception," *Psychological Bulletin*, vol. 129, no. 1, pp. 74–118, 2003, doi: [10.1037/0033-2909.129.1.74](https://doi.org/10.1037/0033-2909.129.1.74).
- [3] A. Vrij, "Criteria-Based Content Analysis: A Qualitative Review of the First 37 Studies," *Psychology, Public Policy, and Law*, vol. 11, no. 1, pp. 3–41, 2005, doi: [10.1037/1076-8971.11.1.3](https://doi.org/10.1037/1076-8971.11.1.3).
- [4] M. L. Newman, J. W. Pennebaker, D. S. Berry, and J. M. Richards, "Lying Words: Predicting Deception From Linguistic Styles," *Personality and Social Psychology Bulletin*, vol. 29, no. 5, pp. 665–675, 2003, doi: [10.1177/0146167203029005010](https://doi.org/10.1177/0146167203029005010).
- [5] L. Zhou, J. K. Burgoon, J. F. Nunamaker, and D. Twitchell, "Automating Linguistics-Based Cues for Detecting Deception in Text-Based Asynchronous Computer-Mediated Communications," *Group Decision and Negotiation*, vol. 13, no. 1, pp. 81–106, 2004, doi: [10.1023/B:GRUP.0000011944.62889.6f](https://doi.org/10.1023/B:GRUP.0000011944.62889.6f).
- [6] A. Vrij, R. Fisher, S. Mann, and S. Leal, "Detecting Deception by Manipulating Cognitive Load," *Trends in Cognitive Sciences*, vol. 10, no. 4, pp. 141–142, 2006, doi: [10.1016/j.tics.2006.02.003](https://doi.org/10.1016/j.tics.2006.02.003).
- [7] J. Cohen, "A Coefficient of Agreement for Nominal Scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960, doi: [10.1177/001316446002000104](https://doi.org/10.1177/001316446002000104).
- [8] J. R. Landis and G. G. Koch, "The Measurement of Observer Agreement for Categorical Data," *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977, doi: [10.2307/2529310](https://doi.org/10.2307/2529310).
- [9] K. Krippendorff, "Reliability in Content Analysis: Some Common Misconceptions and Recommendations," *Human Communication Research*, vol. 30, no. 3, pp. 411–433, 2004, doi: [10.1111/j.1468-2958.2004.tb00738.x](https://doi.org/10.1111/j.1468-2958.2004.tb00738.x).
- [10] F. Al-Hawari and H. Barham, "A Machine Learning Based Help Desk System for IT Service Management," *Journal of King Saud University - Computer and Information Sciences*, vol. 33, no. 6, pp. 702–718, 2021, doi: [10.1016/j.jksuci.2019.04.001](https://doi.org/10.1016/j.jksuci.2019.04.001).
- [11] Q. Li et al., "A Survey on Text Classification: From Shallow to Deep Learning," *arXiv preprint arXiv:2008.00364*, 2020.
- [12] M. Spence, "Job Market Signaling," *The Quarterly Journal of Economics*, vol. 87, no. 3, pp. 355–374, 1973, doi: [10.2307/1882010](https://doi.org/10.2307/1882010).
- [13] G. A. Akerlof, "The Market for "Lemons": Quality Uncertainty and the Market Mechanism," *The Quarterly Journal of Economics*, vol. 84, no. 3, pp. 488–500, 1970, doi: [10.2307/1879431](https://doi.org/10.2307/1879431).
- [14] P. M. Podsakoff, S. B. MacKenzie, and N. P. Podsakoff, "Sources of Method Bias in Social Science Research and Recommendations on How to Control It," *Annual Review of Psychology*, vol. 63, pp. 539–569, 2012, doi: [10.1146/annurev-psych-120710-100452](https://doi.org/10.1146/annurev-psych-120710-100452).
- [15] J. W. Meyer and B. Rowan, "Institutionalized Organizations: Formal Structure as Myth and Ceremony," *American Journal of Sociology*, vol. 83, no. 2, pp. 340–363, 1977, doi: [10.1086/226550](https://doi.org/10.1086/226550).
- [16] M. Blaze, "Rights Amplification in Master-Keyed Mechanical Locks," *IEEE Security & Privacy*, vol. 1, no. 2, pp. 24–32, 2003, doi: [10.1109/MSECP.2003.1193210](https://doi.org/10.1109/MSECP.2003.1193210).
- [17] M. Hanke and R. Sabel, "One Tile, Five Categories: A Coding Taxonomy of Campus Suggestion-Box Submissions Against What the Engagement Pulse Tile Carries Forward," *Slop University technical report*, 2026, doi: [10.5555/slop.tlmg6a](https://doi.org/10.5555/slop.tlmg6a).
- [18] S. Okwuosa and T. Solberg, "Reserved, Not Shortened: Designing and Evaluating a Paper-Slip Reservation Scheme for the Shared Staff Microwave," *Slop University technical report*, 2026, doi: [10.5555/slop.yvfijy](https://doi.org/10.5555/slop.yvfijy).
- [19] D. R. Cox, "Regression Models and Life-Tables," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 34, no. 2, pp. 187–202, 1972, doi: [10.1111/j.2517-6161.1972.tb00899.x](https://doi.org/10.1111/j.2517-6161.1972.tb00899.x).